

5 Safety recommendations

5.1 Targeted audience

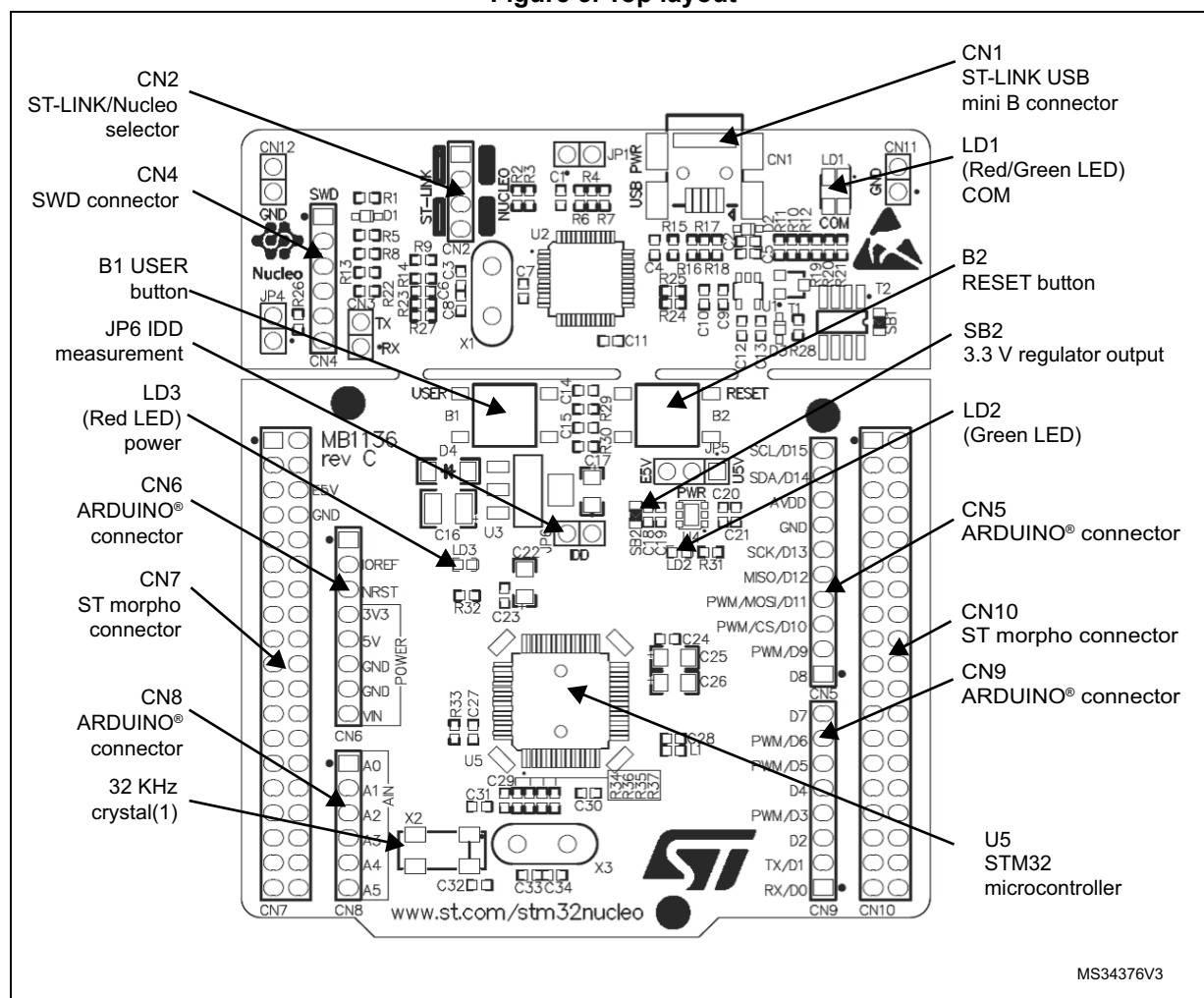
This product targets users with at least basic electronics or embedded software development knowledge such as engineers, technicians, or students. This board is not a toy and is not suited for use by children.

5.2 Handling the board

This product contains a bare printed circuit board and like all products of this type, the user must be careful about the following points:

- The connection pins on the board might be sharp. Be careful when handling the board to avoid hurting yourself.
- This board contains static-sensitive devices. To avoid damaging it, handle the board in an ESD-proof environment.
- While powered, do not touch the electric connections on the board with your fingers or anything conductive. The board operates at a voltage level that is not dangerous, but components might be damaged when shorted.
- Do not put any liquid on the board and avoid operating the board close to water or at a high humidity level.
- Do not operate the board if it is dirty or dusty.

Figure 3. Top layout



1. Crystal might be present or not depending on the board version. Refer to [Section 7.9.2](#).

7.4.1 Driver

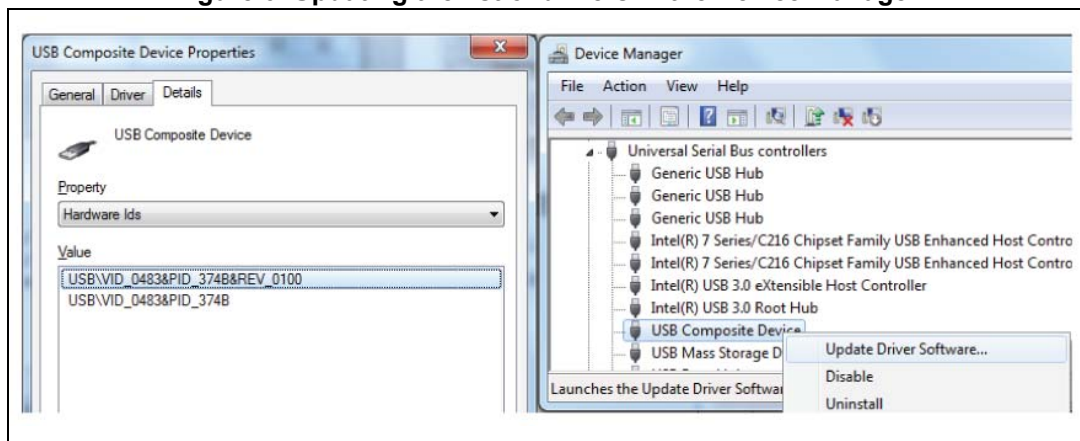
Before connecting the Nucleo-64 board to a Windows® PC via USB, a driver for ST-LINK/V2-1 must be installed. It can be downloaded from the www.st.com website.

In case the STM32 Nucleo-64 board is connected to the PC before installing the driver, the PC device manager might report some Nucleo interfaces as “Unknown”.

To recover from this situation, after installing the dedicated driver, the association of “Unknown” USB devices found on the STM32 Nucleo-64 board to this dedicated driver, must be updated in the device manager manually.

Note: It is recommended to proceed using a USB Composite Device, as shown in [Figure 6](#).

Figure 6. Updating the list of drivers in the Device Manager



7.4.2 ST-LINK/V2-1 firmware upgrade

The ST-LINK/V2-1 embeds a firmware upgrade mechanism for the in-situ upgrade through the USB port. As the firmware might evolve during the lifetime of the ST-LINK/V2-1 product (for example new functionality, bug fixes, support for new microcontroller families), it is recommended to visit www.st.com website before starting to use the STM32 Nucleo board and periodically, to stay updated with the latest firmware version.

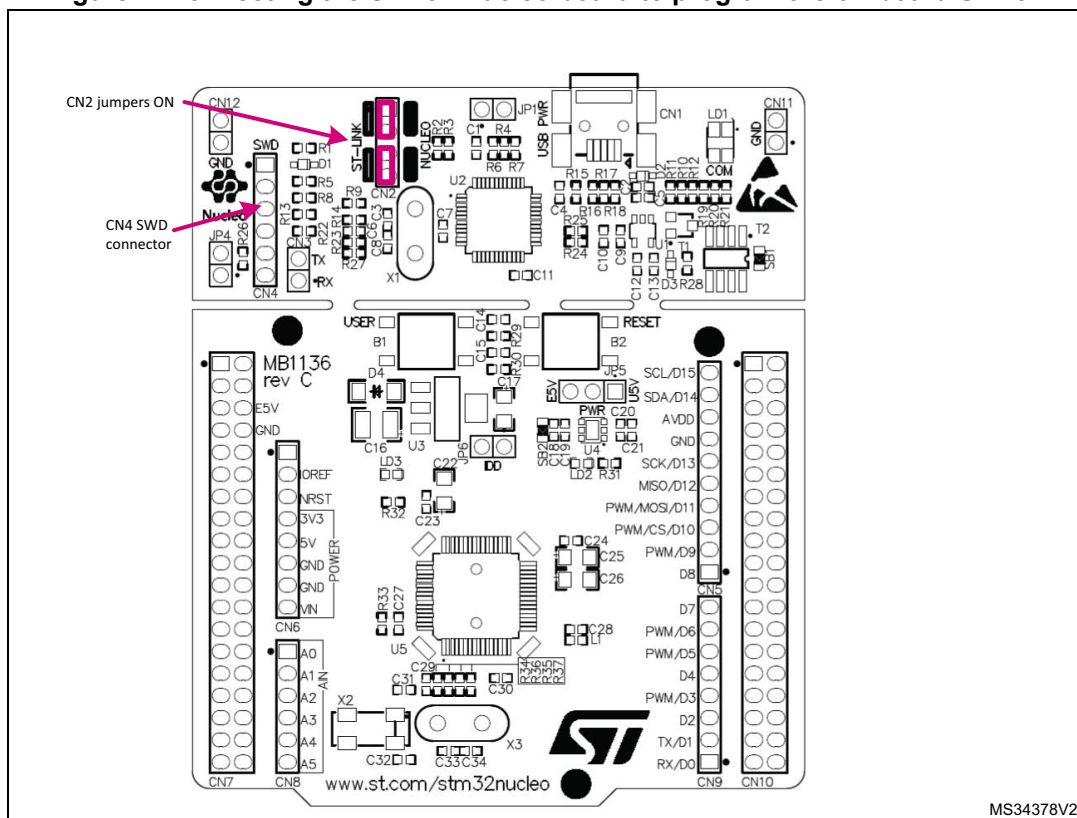
7.4.3 Mass storage interface

The ST-LINK/V2-1 provides an optional mass storage interface, allowing the user to program a target application (binary or hex file) through a virtual mass storage disk. The availability of this interface can be selected during the firmware upgrade, as detailed in the technical note *Overview of ST-LINK derivatives* (TN1235). Make sure that the host PC system (including antivirus applications) allows the mass storage disk before activating the interface.

7.4.4 Using the ST-LINK/V2-1 to program and debug the STM32 on board

To program the STM32 on the board, plug in the two jumpers on CN2, as shown in red in [Figure 7](#). Do not use the CN4 connector as this could disturb communication with the STM32 microcontroller of the STM32 Nucleo board.

Figure 7. Connecting the STM32 Nucleo board to program the on-board STM32



MS34378V2

7.4.5 Using ST-LINK/V2-1 to program and debug an external STM32 application

It is very easy to use the ST-LINK/V2-1 to program the STM32 on an external application. Simply remove the two jumpers from CN2 as illustrated in [Figure 8: Using ST-LINK/V2-1 to program the STM32 on an external application](#), and connect the application to the CN4 debug connector according to [Table 5](#).

Note: SB12 NRST (target STM32 RESET) must be OFF if CN4 pin 5 is used in the external application.